

Abstract

This case study investigates popularity-biased feedback loops in online grocery recommendation systems, specifically on the Instacart platform. While the phenomenon of algorithmic reinforcement is well documented in music and e-commerce, its implications for essential consumer goods remain under-researched. This paper explores whether ranking algorithms like frequently bought together and personalized search-ranking create a cycle that favors globally popular products while systematically marginalizing niche and local brands.

By analyzing user order histories, the study tests for increasing frequencies of high-popularity items over time to identify feedback-loop exposure in the absence of direct exposure data. Utilizing aggregation metrics including mean log popularity and the Herfindahl-Hirschman Index, the research quantifies changes in product diversity from both consumer and producer perspectives. The findings suggest that these feedback loops may reduce consumer choice by limiting exposure to food varieties and create invisible barriers for newcomers in the digital grocery market, potentially distorting market competition and consumer nutrition.

1 Introduction

Products on online grocery shopping sites are becoming increasingly visible through recommendation algorithms, “frequently bought together” features, and personalized search-ranking schemes, all intended to increase frequency of purchase. When these ranking systems evaluate products based on how often they were historically purchased, the system sets up a positive feedback loop that encourages consumers to purchase the same goods more in the future, resulting in increased popularity scores. In literature, there exists ample evidence about such feedback loops in music streaming (Celma 2010), video recommendation (Chaney et al. 2018), and e-commerce (Joachims et al. 2005). However, this problem has not been as well-researched in the online grocery space. From the consumer perspective, limiting product diversity exposure reduces general choice in food types of varieties. On the producer side, feedback loops that favor existing players can create an invisible barrier for niche brands, local producers, and general newcomers to the grocery market.

Our research question is: Does popularity-biased recommendation create feedback loops that increasingly favor popular products while systematically crowding out niche preferences over time? We test this by testing whether users’ order histories show increasing frequencies in globally popular products as they continue to order more, a behavior consistent with feedback-loop exposure in the absence of recommendation logs.

2 Related Work

There is a lot of research surrounding recommender-systems and the idea that recommendation algorithms can create feedback loops that reinforce popularity over time. Chaney et al. (2018) demonstrate that even relatively small amounts of popularity bias in collaborative filtering systems can compound through repeated recommendation cycles, which would eventually reduce the diversity of content users encounter.

Their work shows that when algorithms repeatedly prioritize items that already receive high engagement, those items become increasingly dominant while niche or less popular items can be gradually pushed out of visibility. Another popular work, Mansoury et al. (2020), shows that repeated recommendation rounds amplify exposure inequality, with top-ranked items gaining disproportionate visibility over time.

Together, these studies bring concern that recommender systems do not simply reflect user preferences, but actively shape them through recursive popularity reinforcement.

This is the same pattern that we discussed in the Spotify case study from class. Dinnissen and Bauer (2022) examine how music recommendation systems can influence listening behavior by repeatedly exposing users to already-popular artists and songs. This shows that feedback-loop effects in music streaming emerge surprisingly quickly. In platforms like Spotify, recommendation algorithms continuously learn from prior listening behavior while simultaneously shaping future listening patterns. As a result, mainstream music receives even greater exposure, while niche artists struggle to gain visibility. Our project uses this same feedback-loop framework to groceries.

Compared to entertainment recommendation systems, grocery recommendation research remains relatively limited. Burke (2017) introduces the concept of multistakeholder fairness which argues that recommender systems impact both consumers and producers simultaneously. On the consumer side, fairness involves exposing users to diverse and relevant products rather than only the most popular options. On the producer side, fairness concerns whether smaller brands and niche products are given meaningful opportunities for visibility. However, there is still relatively little research studying these dynamics in online grocery environments. The Instacart dataset has mostly been used for prediction and basket-completion tasks, not for analyzing longitudinal popularity concentration or feedback-loop behavior.

3 Data and Problem Formulation

3.1 Dataset

We use a publicly available Instacart prior-order dataset, archived on Kaggle after Instacart had previously posted it as part of a data science competition. The dataset contains the complete recorded purchase histories of 206,209 anonymized users across 3,214,874 user-orders. Each observation includes a user identifier, an order sequence number reflecting chronological order placement, and a list of products purchased. The dataset spans orders 1 through 100+, but the sample size shrinks substantially beyond order 20 (from 206,209 users at order 1 to 50,731 at order 20), so we treat orders 1-20 as the primary analysis window where statistical power is highest.

3.2 Popularity Segmentation

We compute a global purchase count for each product across the entire dataset. We define two popularity tiers:

- *Popular*: top decile by global purchase count (threshold: $\geq 1,021$ purchases; $\sim 49,000$ unique products qualify)
- *Ultra-popular*: top percentile (threshold: $\geq 9,931$ purchases; ~ 490 products)

3.3 Outcome Metrics

For each order index t across users, we compute three basket-level outcome metrics:

1. **Popular-item share $P(t)$** : fraction of items in the basket belonging to the popular tier.
2. **Ultra-popular-item share $U(t)$** : fraction belonging to the ultra-popular tier.
3. **Mean log global popularity $L(t)$** : average of $\log(\text{global purchase count})$ across all items in the basket.
4. **Within-basket HHI $H(t)$** : Herfindahl-Hirschman Index over product-type shares within the basket, measuring internal basket concentration (lower = more diverse).

We then compute the cross-user mean of each metric at each order index, yielding a trend curve over order history.

3.4 User-Level First-vs-Last Analysis

To complement aggregate trends, we compare each user's first and most recent orders by calculating the individual change in their popular-item share ($\Delta P = P(\text{last}) - P(\text{first})$). This approach reveals individual variations (heterogeneity) that are often hidden by population averages.

4 Results

4.1 Popular-Item Concentration Increases Over Order History

Table 1 and Figure 1 (described below) show the evolution of mean popular-item share across order indices 1-20.

Metric	Order 1	Order 20	Slope (per order)
Popular-item share	0.792	0.819	+0.00146
Ultra-popular share	0.400	0.435	+0.00204
Mean log global popularity	8.617	8.810	+0.01075
Within-basket HHI	0.200	0.192	-0.00069

Table 1. Key trend statistics over orders 1-20.

All three popularity-concentration metrics trend upward over the first 20 orders. Popular-item share rises by 2.7 percentage points, an increase of approximately 3.4% relative to baseline. Ultra-popular share rises by 3.5 percentage points (8.8% relative), suggesting that the most extreme concentration effect is occurring at the very top of the popularity distribution. Mean log global popularity, a continuous measure that does not depend on threshold choice, increases by 0.193 log-units, confirming the pattern independently of any tier cutoff. The within-basket HHI trend moves in the expected direction: as carts become more concentrated in a narrow set of popular products, within-basket diversity falls from 0.200 to 0.192 over the first 20 orders.

Although we are only examining the difference in the order 1 and 20, the popular-item share curve from each order increases through orders 5-15 at approximately linear rate as it reaches 81.9% at order 20. Additionally, there is no evident reversal or plateau in the primary analysis window. For the Ultra-popular share by order index, the top-1% products rises more steeply: from 40.0% at order 1 to 43.5% at order 20. The slope per order exceeds the broad popular-share metric, implying that feedback-loop dynamics are especially pronounced for the most globally dominant products.

4.2 Not All Users Move in the Same Direction

The aggregate trends in Section 4.1 are consistent with a feedback loop narrative, but user-level heterogeneity is significant to examine.

Percentile	ΔP
10th	-0.265
25th	-0.111
50th (median)	0.000
75th	+0.111
90th	+0.261
Mean	-0.00149

Table 2. Distribution of user-level first-vs-last change in popular-item share (ΔP).

The median user shows no change in popular-item share between their first and most recent order. The mean shift is slightly negative (-0.00149), which may appear to contradict the positive aggregate trend. This apparent discrepancy arises from survivorship composition: orders 1-20 aggregate over progressively smaller, self-selected subpopulations of users. Users who accumulate more orders (and thus appear at higher order indices) may differ systematically from one-time or low-frequency shoppers. The aggregate trend reflects compositional change across the user population at each order index, not necessarily within-user drift for the median user.

Interestingly, the 10th-to-90th percentile range spans from -0.265 to +0.261, indicating that roughly one-fifth of users shift by more than ± 25 percentage points. This level of heterogeneity strongly motivates future stratified analyses (by user tenure, basket size, product category) to identify which user subgroups are most exposed to concentration effects.

4.3 Trend Robustness Beyond Order 20

The concentration patterns established in the first 20 orders persist and even intensify at longer order horizons. Popular-item share continues rising through order 50 (~83.3%) and stabilizes in the 83-84% range beyond that, higher than at any point in the first 20 orders. The leveling off may reflect a behavioral steady state or increasing measurement noise as sample sizes decline. The HHI curve similarly stabilizes but at a lower level than its order-1 value, consistent with a persistent reduction in within-basket diversity.

5 Modeling Approach and Shortcomings

5.1 Observational Design

Our analysis is entirely observational. We observe what users purchased and when, but due to a lack of data, we have no access to recommendation and search query logs, i.e. what users actually see and what they searched in the process of buying their groceries.

This means we observe behavioral signatures aligned with feedback loops but not legitimate casual evidence of their existence. We came up with a few alternative explanations for our observed trends in popular-item share:

1. New users may start with the intention of buying new items, then converge to preferred staple

products, which happen to be globally popular and potentially cheaper, as they learn their preferences. This would produce an upward trend in popular-item share through preference learning alone, without any actual algorithmic reinforcement.

2. It could be the case that users' grocery needs change over time. Young adults may add popular household staples as their household matures. Since the dataset is anonymized, we cannot sufficiently control for household composition.
3. Users who continue to shop on Instacart over 20+ orders may differ from those who churn, perhaps by being more mainstream consumers that naturally prefer popular products. If users who prefer niche products disproportionately churn, the cohort of users measured at our high order indices will subsequently appear more popular-concentrated.

5.2 Metric Limitations

Within-order product-type uniqueness ratio was uninformative in this dataset (constant at 1.0 across all orders and users), reflecting the fact that users virtually never purchase duplicate items within a single order. We therefore pivoted to the HHI and global popularity metrics described above. The HHI captures within-basket category concentration rather than item-level uniqueness, which is a related but distinct construct.

6 Discussion and Recommendations

6.1 Relation to Feedback Loops

The trends in Section 4 are best understood through the lens of algorithmic feedback loops. A feedback loop in a recommender system arises when the outputs of the system become inputs to the next round of training data. When a popularity-biased ranker chooses product A over product B, users are more likely to purchase A, which inflates A's purchase count. This in turn strengthens product A's ranking in the next cycle. Over time, this self-reinforcing produces an increasing divergence between market leaders and new market products. Thus, the underlying quality or preference match of the two products might be comparable, however they are recommended at different levels.

Our observational evidence is consistent with this mechanism. The upward trend in popular-item share across the first 20 orders is precisely what a feedback loop predicts: each successive order is placed in an environment where popular products have accumulated even more historical signal than they had at the previous order. The ultra-popular items further suggests the compounding effect is most intense at the very top of the distribution.

Furthermore, it is important to note that the emergence of such an algorithm does not require explicit intent from platform developers. Platform designers who optimize recommendation for near-term conversion will naturally surface popular products more often, because those products have more historical evidence of purchase. No explicit popularity bias policy is required for the loop to operate. It emerges as a structural consequence of using purchase frequency as a ranking signal without corrective mechanisms.

6.2 Relation to Individual Fairness

In this study, individual fairness is directly related to consumer welfare. Popularity-biased recommendations create a significant disparity in individual welfare, especially for users whose preferences lie in the long tail of the distribution. For consumers with niche dietary needs, cultural

preferences, or specialized tastes, the algorithmic behavior that crowds out niche preferences can be considered a form of representation harm. As the recommender system prioritizes higher rates of conversion via globally popular items, it fails to represent the actual diversity of the underlying product space. Over time, niche-preference users face a crossroads. They either have to search harder to find products that the algorithm has systematically buried, or abandon their niche tastes and conform to globally popular tastes. The system performance therefore falls short of Pareto optimality, as the gains in convenience for mainstream consumers are achieved at the direct expense of discovery and utility for niche-preference consumers.

The connection to feedback loops deepens this individual fairness concern. Each time the system steers a niche-preference user toward a popular product, it records that purchase as a signal that the popular product is appropriate for that user. The user's preference profile is therefore corrupted in the direction of popularity. This makes it increasingly unlikely that the algorithm will recommend users the niche alternatives the user would have preferred. The user-level heterogeneity we document in Table 2 is consistent with this dynamic: the wide interquartile range (-0.111 to $+0.111$) suggests that some users resist the drift while others are substantially reshaped by it, with the 10th-to-90th percentile spread of roughly ± 26 percentage points indicating that the individuals most harmed are harmed significantly. Under standard individual fairness frameworks, a system that treats two users with similar preferences differently fails the fundamental requirement that similar individuals receive similar treatment.

6.3 Relation to Group Fairness

Moving to the perspective of group fairness, we separate products into ultra-popular and niche brands. Our aforementioned analysis suggests that the recommender functions as a structural (invisible) barrier to entry for small-scale and local producers. A feedback loop that reinforces existing purchases creates this rich-get-richer effect, putting new or niche entrants at risk as they lack the historical transaction volume to trigger the recommendation threshold. This pattern of market concentration is empirically backed up by the trends in HHI and the decline in within-basket diversity. As consumers continue to converge on a narrowing product set, the marketplace fails to provide equitable exposure for the bottom 90% of the catalog.

The feedback loop mechanism is what transforms this from a static disadvantage into a compounding one. A new or niche producer who enters the platform at any point in time is competing against established market leaders whose popularity signals have been accumulating across millions of transactions. Even if the new product is of equal or superior quality, it begins each recommendation cycle with a near-zero purchase history, placing it at the bottom of any frequency-weighted ranking. Each ordering cycle in which it is not recommended is a cycle in which it accumulates no additional signal, while popular incumbents continue to widen their lead. The group-level consequence from captured in the ultra-popular group is that a small set of already-dominant products captures an ever-larger share of basket composition across the entire user population. This is not an outcome that reflects consumer preference formation alone, rather it is in part an artifact of a system design that systematically favors products with historical momentum over products with potential.

6.4 Countermeasures & Future Work

If we can confirm this feedback loop hypothesis through formal causal analysis, there are a number of interventions that online grocery platforms can take. These include but are not limited to: periodically recommending niche items into recommendation to interrupt reinforcing cycles, apply a saturation

function to popularity signals before using them in ranking, set minimum exposure quotas for items outside the top popularity decile, and/or post-process recommendation lists to ensure a minimum fraction of niche items.

The most evident limitation of this work is the absence of exposure data. To properly establish causality requires a natural experiment, simulation studies that test the feedback loop mechanism under different ranking policies, or methods involving instrumental variables (increasingly popular in economics). The most realistic option if we were to work more on this in the future would be a simulation, comparing neutral, popularity-biased, and diversity-aware ranking policies on hypothetical user trajectories, calibrated to the Instacart distribution. Another thing we would add in the future would be decomposing the aggregate trend by user tenure (orders 1-5 vs 6-20), basket size, department, and geography, assuming they are publicly available. Looking back at Table 2, subgroup-specific dynamics may differ substantially from the population-level average.

7 Conclusion

Using the Instacart prior-order dataset, it is evident that the concentration of globally popular products observes a clear and consistent upward trend over time. The share of popular (top-10%) products rises by 2.7% over the first 20 orders, ultra-popular (top-1%) products by 3.5%, and mean log global product popularity rises monotonically through the same window. These patterns are aligned with the presence of a popularity-reinforcing feedback loop. While we cannot go as far as establishing causality from purchase data alone, the behavior we observe is consistent with simulation and empirical findings from published music, video, and e-commerce recommender research.

The implications for both individual and group fairness are clear. Consumers may experience a progressive reduction of their effective set of grocery choices, and small/niche producers may face algorithmic disadvantage that develops over the platform’s lifetime. In order to address these dynamics, we would recommend a range of approaches used in similar fields (music, video, e-commerce) such as exploration-based injection, popularity dampening, exposure quotas, and diversity-aware re-ranking. We hope this empirical analysis leads to both further causal investigation and practical algorithm auditing in the online grocery space.

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